

Assignment 5: Unconsidered Metrics (SpO₂)

SpO₂ (Peripheral Oxygen Saturation) is a non-invasive measurement of the percentage of hemoglobin in arterial blood that is saturated with oxygen. It is commonly measured using pulse oximetry. Pulse Oximetry is where a sensor placed on the finger, ear, or toe estimates arterial oxygen saturation through light absorption. SpO₂ reflects the effectiveness of oxygen transport in the bloodstream and provides an indirect assessment of pulmonary oxygenation.

Clinically, normal SpO₂ values typically range from 95% to 100% in healthy individuals breathing room air. Values between 90% and 94% may indicate mild hypoxemia, while values below 90% are considered significant hypoxemia and are associated with impaired tissue oxygen delivery. Critically low SpO₂ levels, particularly below 85%, may lead to cellular damage and increased mortality. However, acceptable target ranges may differ in patients with chronic respiratory conditions where lower oxygen saturation targets are sometimes clinically appropriate.

For triaging, SpO₂ is an important indicator of respiratory and circulatory status because it helps identify patients with inadequate oxygenation who may require urgent intervention. Reduced SpO₂ can indicate conditions such as respiratory failure and airway obstruction. In emergency and critical care settings, SpO₂ is used alongside other vital signs such as FiO₂ to assess illness severity and prioritize treatment. Although SpO₂ is highly useful for rapid bedside assessment, it must be interpreted within the broader clinical context because factors such as carbon monoxide poisoning and anemia may affect measurement accuracy.